

Title: Methods protocol for counterfactual life table analysis of excess mortality

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Summary

We calculate life tables and associated life expectancies by sex for a selection of 34 countries for the years 2000 through 2024. For the years 2020 through 2024 we calculate counterfactual life expectancies based on the continuation of pre-pandemic mortality trends and derive associated life expectancy deficits – the difference between actual and counterfactual life expectancy – as an age standardized and trend adjusted measure of excess mortality.

The data basis for our calculation are midyear population and death counts by age, sex, and country sourced from the Short Term Mortality Fluctuations Database (HMD-STMf [1]), the Human Mortality Database (HMD, [2, 3]), the UN World Population Prospects (WPP-2024 [4]), and national statistical offices (ONS [5], CDC [6]). For a complete list of sources by country see methods appendix “Data sources”.

Age specific death counts are harmonized to a single year age grouping with open age group 100+ using the PCLM method [7, 8]. Weekly death counts are aggregated to annual counts and midyear population estimates are converted to person years of exposure adjusting for the effect of leap weeks. To adjust for systematic bias in registered deaths between annualized STMf data and HMD annual totals we estimate correction factors based on a statistical model (see methods appendix “Bias correction of age specific annual death counts”).

Given the harmonized annual death counts and population exposures by age and sex we calculate period life tables using the standard piecewise-exponential model. We then fit a Poisson-Lee-Carter model [9] using the R library StMoMo [10] to the available pre-pandemic data since 2000 and forecast counterfactual age specific mortality rates and death counts over the years 2020 through 2024. Life expectancy deficits (LED) are then derived along with P-scores [11] and the Mean unfulfilled lifespan (MUL, [12]) as alternative measures of excess. We decompose the life expectancy deficits into age specific contributions using the Arriaga method [13].

Throughout the analysis population exposures and annual death counts by sex are aggregated as needed to derive results for both sexes combined and for the joint period 2020-2024.

We perform simulation based uncertainty propagation throughout our analysis via a Poisson parametric bootstrap of the life table estimates and a random walk simulation of the Lee-Carter forecast. Based on this we derive a p-value for the estimated life expectancy deficit, i.e. the probability of observing or exceeding an estimated life expectancy deficit given the continuation of pre-pandemic trends.

The complete analysis workflow is explained in detail in the methods appendix. The analysis script is available from <https://github.com/jschoeley/e0deficit>.

Data sources¹

The basis for our analysis are death and population counts by age, year, sex, and country. We source the death counts from:

- The Short-term mortality fluctuations (HMD-STMF, [1]) input data base for weekly death counts by age and sex across countries. This is our main source. The data is updated weekly and includes the most recent information on death counts. Using HMD-STMF allows for timely updates of annual life expectancy estimates across countries.
- The UK's Office for National Statistics (ONS, [5]) annual death counts by age and sex for England and Wales. We use this data for years 2010 through 2019 because the age grouping in the HMD-STMF data for these years is too coarse for reliable life expectancy estimates.
- The US Centers for Disease Control and Prevention (CDC, [6]) for annual death counts by age and sex for the United States. We use this data for years 2000 through 2019 for the US, lacking any HMD-STMF data for these years.
- The Human Mortality Database (HMD, [2, 3]) for annual death counts and exposures for those countries for which a complete series is available through 2024.

Population data form the basis for our person year exposure estimates and are sourced from:

- The World Population Prospects 2024 (WPP-2024, [4]) for midyear population estimates and projections by age, year and country.
- The HMD for midyear population estimates by age, year and country for those countries for which a complete series is available through 2024 and for the regions of the UK.

For a list of data sources by country and year see Table 1.

Data harmonization²

Death count age harmonization

As the HMD-STMF input data is not harmonized in terms of age grouping we perform a model based ungrouping of deaths into single ages for each region-year-sex stratum using the Penalized Composite Link Method (PCLM) [7] implemented in the R package `ungroup` [8].

The original death counts over single year age groups $\mathbf{d}^{\text{single}} = (d_{j=1}, \dots, d_J)^\top$ have been grouped (summed) into discrete age bins $\mathbf{d}^{\text{grouped}} = (d_{i=1}, \dots, d_I)^\top$ of varying width. We assume the grouped death counts to arise from a Poisson distribution with mean μ_i :

$$d_i^{\text{grouped}} \sim \text{Poisson}(\mu_i) .$$

The PCLM methods seeks to recover the means of the original single year death counts, $\boldsymbol{\gamma} = (\gamma_{j=1}, \dots, \gamma_J)^\top$. To that end, one can encode the age aggregation into a binary composition matrix

¹github.com/jschoeley/e0deficit/blob/main/src/12-download_death.R

github.com/jschoeley/e0deficit/blob/main/src/11-download_population_and_rates.R

²https://github.com/jschoeley/e0deficit/blob/main/src/21-harmonize_death_with_exposures.R

Table 1: Data sources by country and year.

Country	Period	Death counts	Population
Austria	2000-2024	HMD-STMF	WPP-2024
Belgium	2000-2024	HMD	HMD-EST
Bulgaria	2000-2024	HMD-STMF	WPP-2024
Switzerland	2000-2024	HMD	HMD-EST
Chile	2000-2015	HMD	WPP-2024
Chile	2016-2024	HMD-STMF	WPP-2024
Czech Republic	2000-2004	-	WPP-2024
Czech Republic	2005-2024	HMD-STMF	WPP-2024
Germany	2000-2024	HMD-STMF	WPP-2024
Denmark	2000-2024	HMD	HMD-EST
Estonia	2000-2024	HMD	HMD-EST
Spain	2000-2024	HMD-STMF	WPP-2024
Finland	2000-2024	HMD	HMD-EST
France	2000-2024	HMD-STMF	WPP-2024
England and Wales	2000-2009	-	HMD-EST
England and Wales	2010-2019	ONS	HMD-EST
England and Wales	2020-2022	HMD-STMF	HMD-EST
England and Wales	2023-2024	HMD-STMF	HMD-PRJ
Northern Ireland	2000-2021	HMD	HMD-EST
Northern Ireland	2022	HMD-STMF	HMD-EST
Northern Ireland	2023-2024	HMD-STMF	HMD-PRJ
Scotland	2000-2022	HMD-STMF	HMD-EST
Scotland	2023-2024	HMD-STMF	HMD-PRJ
Greece	2000-2014	HMD	WPP-2024
Greece	2015-2024	HMD-STMF	WPP-2024
Croatia	2000	-	WPP-2024
Croatia	2001-2024	HMD-STMF	WPP-2024
Hungary	2000-2024	HMD-STMF	WPP-2024
Israel	2000-2024	HMD-STMF	WPP-2024
Iceland	2000-2024	HMD-STMF	WPP-2024
Italy	2000-2010	-	WPP-2024
Italy	2011-2024	HMD-STMF	WPP-2024
Japan	2000-2024	HMD	HMD-EST
Lithuania	2000-2024	HMD	HMD-EST
Luxembourg	2000-2024	HMD-STMF	WPP-2024
Latvia	2000-2024	HMD	HMD-EST
Netherlands	2000-2024	HMD-STMF	WPP-2024
Norway	2000-2024	HMD	HMD-EST
Poland	2000-2024	HMD-STMF	WPP-2024
Portugal	2000-2024	HMD	HMD-EST
Sweden	2000-2024	HMD	HMD-EST
Slovenia	2000-2024	HMD-STMF	WPP-2024
Slovakia	2000-2024	HMD	HMD-EST
Taiwan	2000-2024	HMD-STMF	WPP-2024
USA	2000-2019	CDC	WPP-2024
USA	2020-2024	HMD-STMF	WPP-2024

$\mathbf{C}$ with dimensions $I \times J$ such that $\mathbf{d}^{\text{grouped}} = \mathbf{C}\mathbf{d}^{\text{single}}$. The grouped Poisson rates are then connected to the single age means via the composite link

$$\mu = \mathbf{C}\gamma.$$

By specifying that the γ are a smooth function of age,

$$\gamma_j = \exp(s(x(j))),$$

where $s(x(j))$ is a P-spline over the starting age x of single year age groups $j \in 1, \dots, J$, a penalized likelihood solution can be found for the vector of mean death counts by single age. To get an integer estimate of the death count in single age x we round $\gamma_x(j)$ down to the next integer,

$$\hat{d}_x^{\text{single}} = \lfloor \gamma_{x(j)} \rfloor,$$

such that $\hat{d}_x$ corresponds to the mode of the Poisson distribution with rate $\gamma_{x(j)}$.

Because the age grouping scheme may vary within a year, the ungrouping is separately applied to each region $\times$ sex $\times$ year $\times$ age pattern combination in the input data. In a final step, the harmonized sub-annual HMD-STMF death counts are summed into annual death counts by age, $\hat{d}_{xy}$.

Exposure time calculation

We derive person years of exposure by age and year, E_{xy} , from the midyear population estimates, N_{xy}^{midyear} . We choose the HMD midyear population for exposure calculation for those countries where a complete series is available through 2024 and for the regions of UK. Otherwise we choose the World Population Prospects 2024 data which is always complete throughout 2024. To complete HMD exposures for the UK regions we perform a stable population projection based on the most recent HMD population data and 5 year average (2015 through 2019) age specific mortality and fertility rates. Exposures are adjusted for the length of the observation period within a year, taking into account weeks with missing data and leap-weeks.

$$E_{xy} = N_{xy}^{\text{midyear}} n_y^{\text{observed-weeks}} / 52.$$

Bias correction of age specific annual death counts³

The purpose of our bias correction model is to correct any systematic deviations between our age specific mortality rate estimates and the official annual estimates. While the HMD-STMF death counts allow for timely estimates of current mortality they sometimes deviate substantially from official annual estimates due to different population definitions and reporting guidelines in the weekly death reporting compared to the annual reporting. Because these systematic biases are stable in time, they can be estimated and corrected. To that end we estimate a correction factor $\hat{c}_{xy}$ which, if multiplied with the single age annual death counts, $\hat{d}_{xy}$, yields the corrected counts:

$$\hat{d}_{xy}^{\text{corrected}} = \hat{c}_{xy} \cdot \hat{d}_{xy}.$$

We estimate the correction factor via the Poisson model

$$d_{xy}^{\text{benchmark}} \sim \text{Poisson} \left(c_{xy} \cdot \hat{d}_{xy} \right),$$

where the benchmark death counts are taken from the HMD and

$$c_{xy} = \exp(s_y(x|y)),$$

is a smooth age effect estimated separately for each year capturing the time varying deviations between uncorrected and benchmark death counts. The correction factor from the last year with available benchmark data will be used in subsequent years.

³https://github.com/jschoeley/e0deficit/blob/main/src/31-rate_bias_correction.R

Figures 1 and 2 show the estimated correction factors and the bias corrected age specific death counts by year for the example of Austrian females. The model is fit separately to each region and sex.

Figure 1: Demonstration of estimated (red) bias correction factors c_{xy} vs. raw correction factors (black) for Austrian females.

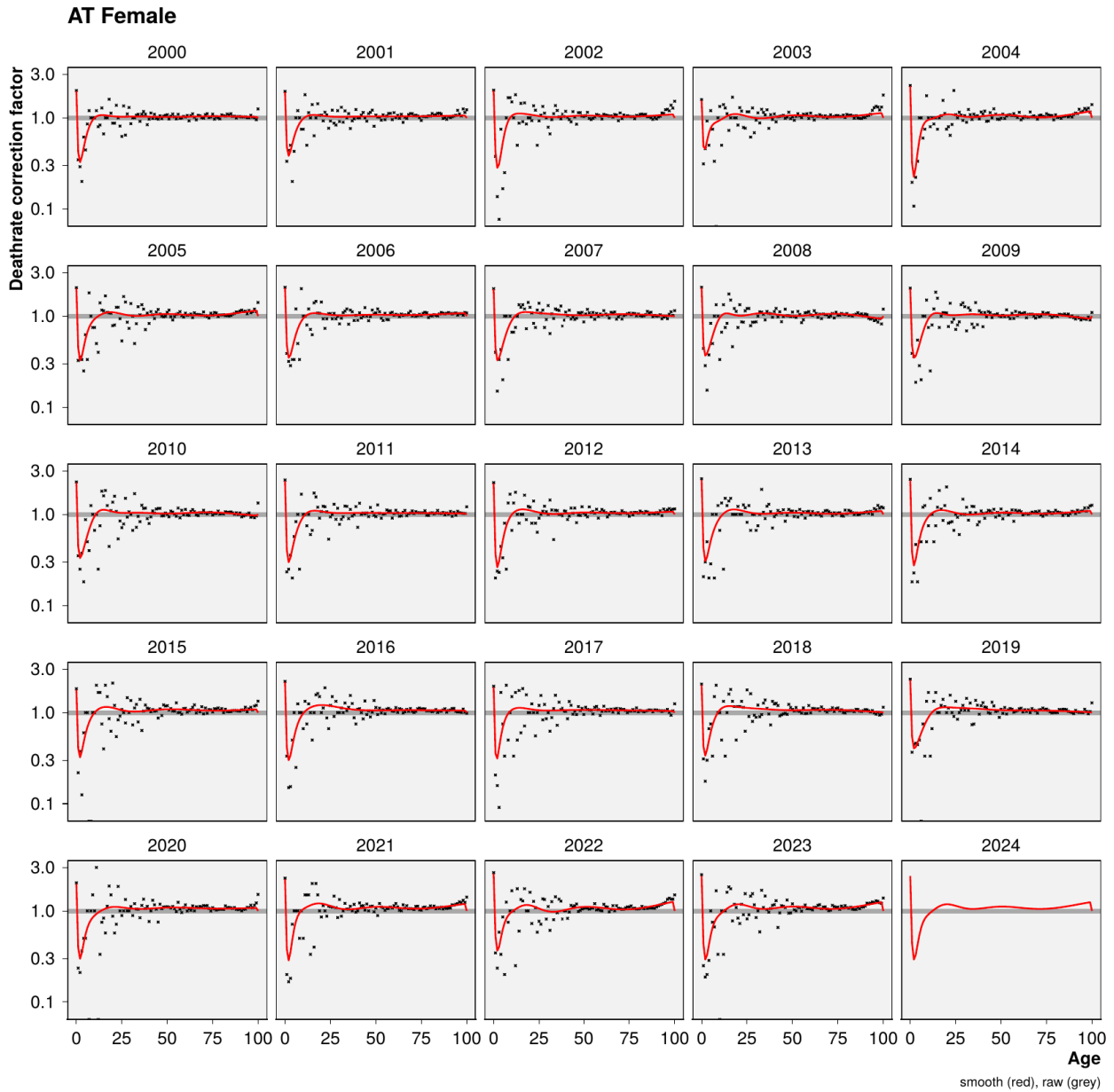
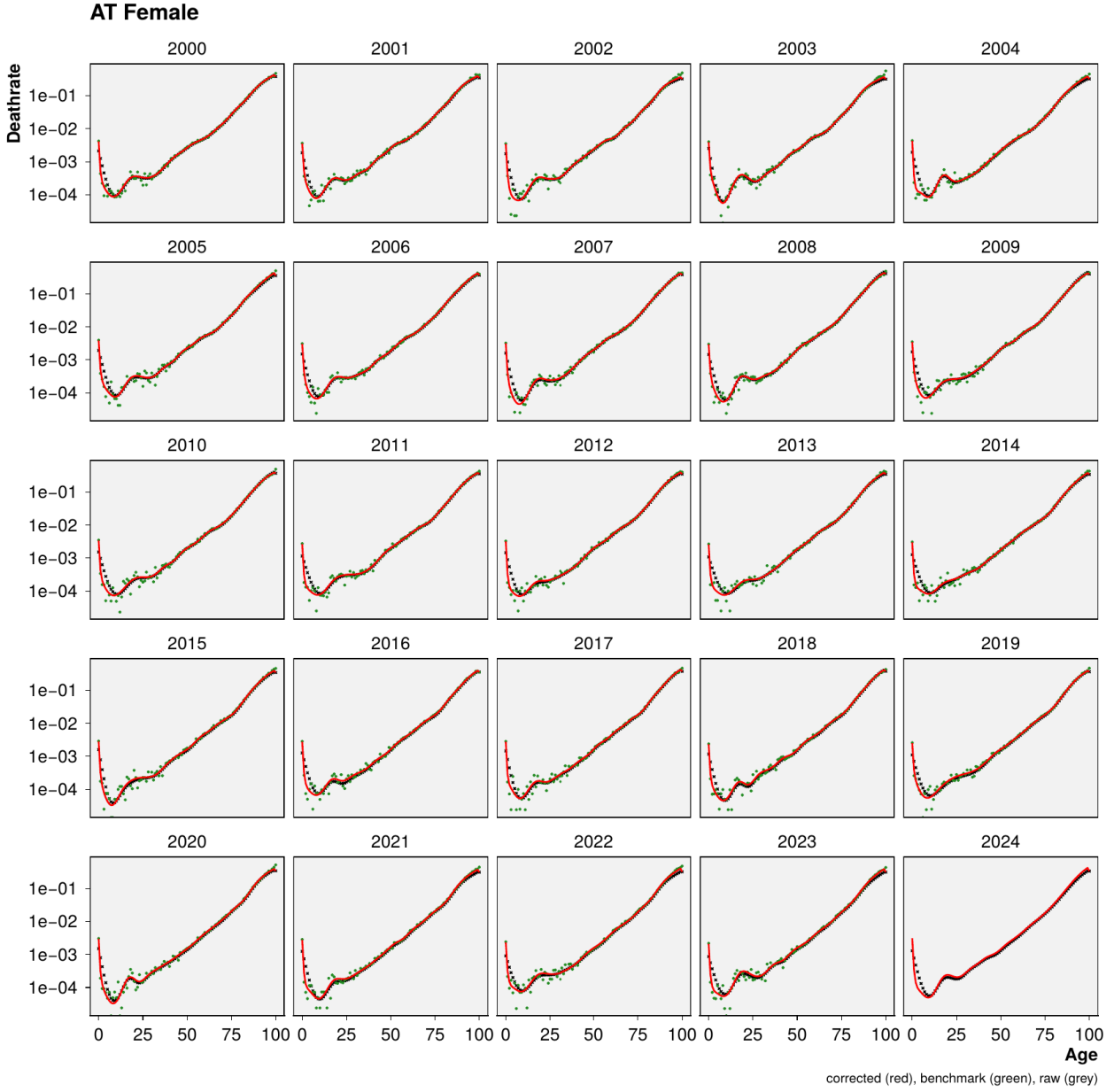


Figure 2: Demonstration of bias corrected ungrouped age specific death rates (red) vs. non-corrected death rates (black) and HMD benchmark death rates (green) for Austrian females.



Counterfactual mortality forecast⁴

In order to calculate the life expectancy under continuation of pre-pandemic trends we forecast age specific mortality rates over the years 2020-2024 using a Poisson-Lee-Carter model given the data available up to January 2020.

We fit a Poisson-Lee-Carter model [9] where death counts at single age x and single year y are distributed

$$d_{xy} \sim \text{Poisson}(\mu_{xy} E_{xy}),$$

⁴https://github.com/jschoeley/e0deficit/blob/main/src/40-counterfactual_projection.R

with death rates

$$\mu_{xy} = \exp(\alpha_x + \beta_x \kappa_y),$$

and identifiability constraints $\sum_y \kappa_y = 0$ and $\sum_x \beta_x = 1$.

We forecast μ_{xy} past the fitting period via the usual random walk with drift over κ_y ,

$$\kappa_{y+1} = \kappa_y + \hat{b} + \text{Normal}(0, \hat{\sigma}).$$

We estimate the variance and mean difference of the estimated year-on-year κ_y changes and simulate a corresponding random walk with drift over the years 2020-2024. The resulting forecasts of κ are then used to derive a corresponding surface of age-period mortality rates $\mu_{xy}^{\text{expected}} = \exp(\alpha_x + \beta_x \kappa_y)$ for forecast horizon $y \in \{2020, 2021, \dots, 2024\}$. We fit the model using the R library StMoMo [10]. Forecast uncertainty is propagated throughout the rest of the analysis via 250 samples of the forecasted $\mu_{xy}^{\text{expected}}$.

Life table inference⁵

We perform a parametric bootstrap of our life table estimates based on 250 Poisson replicates of actual and expected age specific death counts for any population under consideration (for brevity sex, country, and year indices are omitted in this section).

$$d_x^{\text{actual}} \sim \text{Poisson}(\hat{d}_x^{\text{corrected}})$$

$$d_x^{\text{expected}} \sim \text{Poisson}(\mu^{\text{expected}} \cdot E_x)$$

At this point we aggregate the simulated death counts along with the person years of exposure into larger populations of interest, e.g. male and female are summed to total population and annual data are summed to multi-year aggregates. All of the following analysis steps are also performed for the newly created aggregated populations.

For any given population and for each sample draw we calculate the actual and expected central death rates by age as

$$m_x^{\text{actual}} = d_x^{\text{actual}} / E_x,$$

and

$$m_x^{\text{expected}} = d_x^{\text{expected}} / E_x.$$

These rates are then transformed into life tables yielding actual and expected life expectancies, e_x^{actual} and e_x^{expected} , for each population, year, and simulation draw. We write $n_x = 1$ for $x \neq \omega$ and $n_\omega = \infty$ for the width of the age group starting at x with $\omega = 100$ being the open age group closing the life table. Under the piecewise constant assumption we calculate the life tables as

⁵https://github.com/jschoeley/e0deficit/blob/main/src/50-estimation_and_inference.R

$$\begin{aligned}
\Pr(X \geq x + n | X > x) &= {}_n p_x = \begin{cases} 0 & \text{if } x = \omega \\ \exp(-{}_n m_x n_x) & \text{otherwise} \end{cases} \\
\Pr(X > x) &= \ell_x = \begin{cases} 1 & \text{if } x = 0 \\ \prod_{i=0}^{x-1} {}_n p_{x=i} & \text{otherwise} \end{cases} \\
\Pr(x \leq X < x + n) &= {}_n d_x = \begin{cases} \ell_\omega & \text{if } x = \omega \\ \ell_{x+n} - \ell_x & \text{otherwise} \end{cases} \\
\int_x^{x+n} \ell_a da &= {}_n L_x = \begin{cases} 1/{}_n m_x & \text{if } x = \omega \\ {}_n d_x / {}_n m_x & \text{otherwise} \end{cases} \\
\int_x^\infty \ell_a da &= T_x = \sum_{i=x}^\omega {}_n L_{x=i} \\
\mathbb{E}(X | X > x) &= e_x = T_x / \ell_x.
\end{aligned}$$

Now the *life expectancy deficit* can be calculated as the difference between actual and counterfactual life expectancy, the baseline life expectancy under the continuation of pre-pandemic trends,

$$e_0^{\text{deficit}} = e_0^{\text{actual}} - e_0^{\text{expected}}.$$

Note that we define the deficit such that it is negative if actual life expectancy falls short of the expectation, i.e. if there are excess deaths, and positive if there is a life expectancy surplus, i.e. if the expectation is outperformed.

Life expectancy deficits do not necessarily indicate a structural deviation from expected mortality trends as both the expected and the observed life tables are subject to uncertainty. Uncertainty in the expectation is mainly due to forecasting error – we do not know exactly how age specific mortality rates would have developed in the absence of the pandemic. This uncertainty is captured by the random walk of the forecasting model described earlier. Additionally, aleatoric uncertainty in the death counts induces uncertainty in the estimated central death rates which is reflected via our Poisson re-sampling scheme.

Due to the stated sources of randomness one may observe life expectancy deficits even under the continuation of pre-pandemic trends. Thus we perform inference about the estimated e_0^{deficit} by calculating a p-value stating the probability of estimating a life expectancy deficit as large as the realized deficit under the continuation of pre-pandemic trends. Life expectancy deficits under the null hypothesis of pre-pandemic trends are defined as

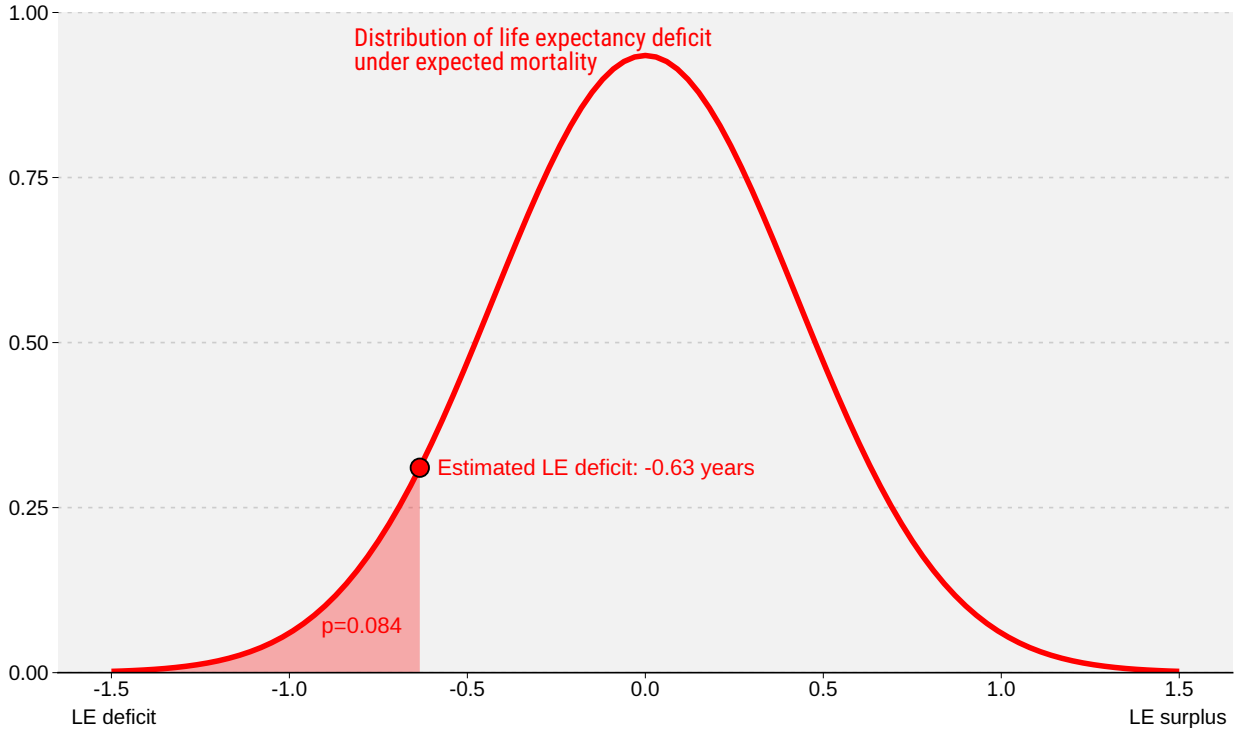
$$T = e_0^{\text{expected}} - \mathbb{E}\left(e_0^{\text{expected}}\right),$$

and we derive the sampling distribution of T from our sample draws of e_0^{expected} . We can then calculate the probability of our calculated e_0^{deficit} given pre-pandemic trends as

$$\Pr\left(e_0^{\text{deficit}} \leq T^*\right).$$

Figure 3 demonstrates the procedure.

Figure 3: Demonstration of p-value calculation for estimated life expectancy deficit.



Age decomposition analysis⁶

We perform the Arriaga decomposition [13] to attribute life expectancy deficits to age specific excess mortality. As the Arriaga decomposition is analytic and can be expressed as a series of simple arithmetic operations it lends itself well to simulation based inference where computation speed is vital. Based on the previously estimated life table quantities we calculate

$${}_n\text{DE}_x = \left({}_nL_x^{\text{actual}} / \ell_x^{\text{actual}} - {}_nL_x^{\text{expected}} / \ell_x^{\text{expected}} \right) \cdot \ell_x^{\text{actual}}$$

$${}_n\text{IE}_x = \left(\ell_x^{\text{expected}} / \ell_x^{\text{actual}} - \ell_{x+n}^{\text{expected}} / \ell_{x+n}^{\text{actual}} \right) \cdot T_{x+n},$$

which yields the contribution of excess mortality at age x to the overall life expectancy deficit:

$${}_n\Delta_x^{e_0} = {}_n\text{DE}_x + {}_n\text{IE}_x.$$

Alternative measures of excess mortality⁷

In addition to the life expectancy deficit we calculate alternative measures of excess mortality as a sensitivity check.

Excess deaths are the difference between actual and expected deaths,

⁶https://github.com/jschoeley/e0deficit/blob/main/src/50-estimation_and_inference.R

⁷https://github.com/jschoeley/e0deficit/blob/main/src/92-sensitivity_check_excess_measures.R

$$\text{Excess}_x = d_x^{\text{actual}} - d_x^{\text{expected}}$$

and can be turned into the P-score[11] by aggregating over age and expressing them as a ratio to total expected deaths,

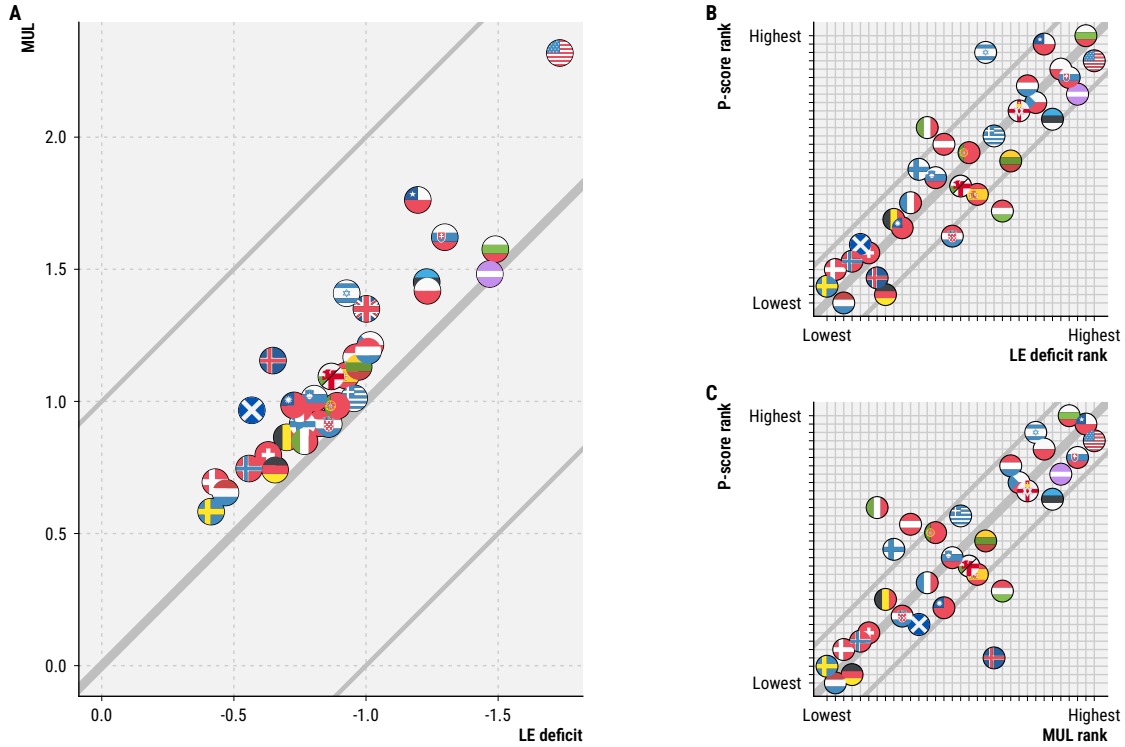
$$\text{Pscore} = \sum_x \frac{\text{Excess}_x}{d_x^{\text{expected}}}.$$

The mean unfulfilled lifespan (MUL, [12]), like the life expectancy deficit, is based on counterfactual life expectancies. MUL is defined as the difference between the average age at death during some period and the average expected age at death under counterfactual mortality conditions.

$$\text{MUL} = \frac{1}{\sum_x d_x^{\text{actual}}} \sum_x \text{Excess}_x \cdot e_x^{\text{expected}}.$$

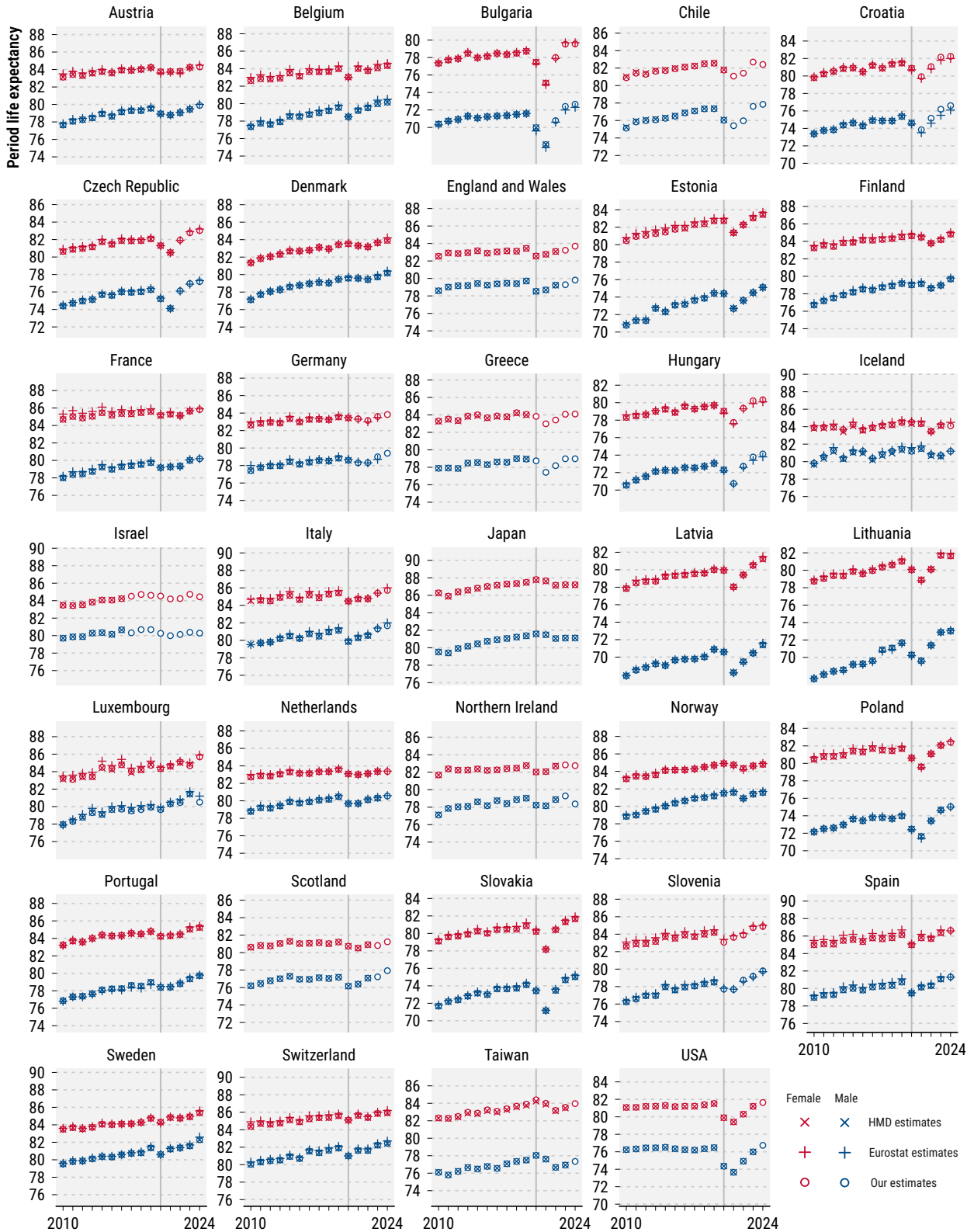
Figure 4 shows a comparison of the three measures with each other across countries.

Figure 4: Cross-country comparison of alternative excess mortality measures for the period 2020-2024. A) Life expectancy deficit vs. mean unfulfilled lifespan, B) Country ranking of life expectancy deficit vs. P-score, C) Country ranking of mean unfulfilled lifespan vs. P-score.



Benchmark tests⁸

Figure 5: Comparison of our life expectancy estimates with past estimates by Eurostat and the Human Mortality Database.



⁸https://github.com/jschoeley/e0deficit/blob/main/src/91-benchmark_check_e0.R

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